THE OBJECTS KILLED BY THE LOCALIZATION FUNCTOR AT A MULTIPLICATIVE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three equivalences, each of which is written out with the morphism it applies the hypotheses to; no step is left to the reader and the closing sentence is a conclusion.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Localization at a multiplicative system. Same chapter and Categories, the construction of the localization and the universal property of the localization functor. Supplies the functor Q and the fact that it carries the morphisms of the multiplicative system to isomorphisms.
- Which morphisms the localization inverts. Categories, Lemma categories-lemma-what-gets-inverted. Supplies the criterion that a morphism becomes an isomorphism in the localization exactly when it can be completed to an element of the multiplicative system, which produces condition (3) from condition (1).

Source reference. Original lemma 05Q9: sources/stacks-categories/pinned-original-source.tex, source lines 5013–5013. Navigation pointer invoked at proof line 2122; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be an additive category. Let S be a multiplicative system. Let X be an object of $\mathcal{C}$. The following are equivalent*

(1) $Q(X) = 0$ *in* $S^{-1}\mathcal{C}$,

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(2) *there exists* $Y \in \mathrm{Ob}(\mathcal{C})$ *such that* $0 : X \to Y$ *is an element of* S*, and*
(3) *there exists* $Z \in \mathrm{Ob}(\mathcal{C})$ *such that* $0 : Z \to X$ *is an element of* S*.*

Proof. If (2) holds we see that $0 = Q(0) : Q(X) \to Q(Y)$ is an isomorphism. In the additive category $S^{-1}\mathcal{C}$ this implies that $Q(X) = 0$. Hence (2) $\Rightarrow$ (1). Similarly, (3) $\Rightarrow$ (1). Suppose that $Q(X) = 0$. This implies that the morphism $f : 0 \to X$ is transformed into an isomorphism in $S^{-1}\mathcal{C}$. Hence by Categories, Lemma 05Q9 there exists a morphism $g : Z \to 0$ such that $fg \in S$. This proves (1) $\Rightarrow$ (3). Similarly, (1) $\Rightarrow$ (2). $\square$